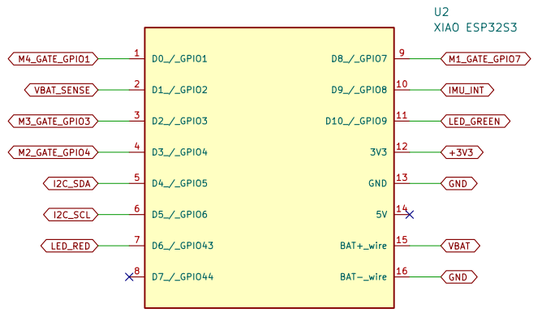
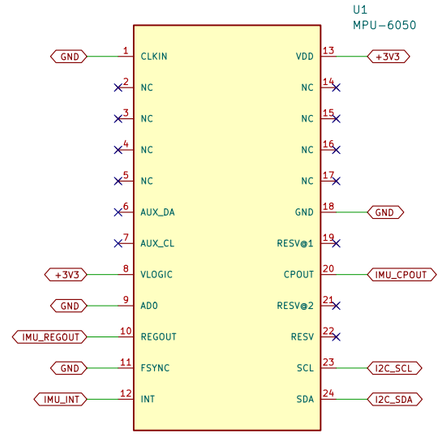
ESP-FLY | Điều khiển, IMU và nguồn

Tái dựng từ Gerber Max Imagination, bản 331d8f50 | 4 lớp | dày 1,2 mm

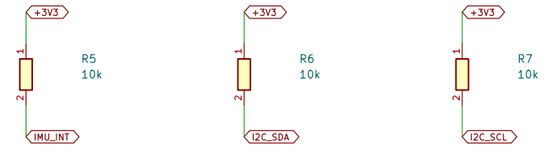
Mô-đun XIAO ESP32S3 và dây pin



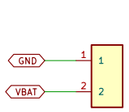
MPU-6050 (ánh xạ chân theo tài liệu nguồn)

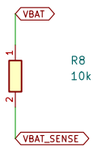


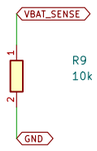
Điện trở kéo lên: INT / SDA / SCL



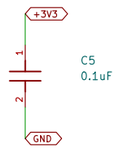
Pin 1S và mạch chia áp

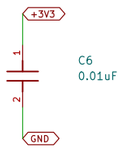


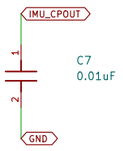


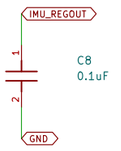


Tụ tách nguồn IMU và bơm điện tích









Nối hai pad BAT+ / BAT- với pad pin ở mặt dưới XIAO bằng hai dây.
VBAT_SENSE đi đến GPIO2 của XIAO trên PCB này. Hãy kiểm tra ánh xạ ADC trong firmware.



Động cơ

Tập tin: Động-cơ.kicad_sch

Đèn LED

Tập tin: Đèn-LED.kicad_sch

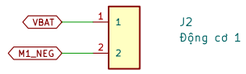
Gerber nguồn 331d8f50; rà soát DRC trước khi chế tạo
Thiết kế gốc: Max Imagination

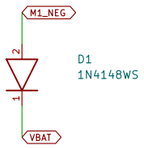
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Title: ESP-FLY | Điều khiển, IMU và nguồn		
Size: A3	Date: 2026-09-25	**Rev: Tái dựng 1**
KiCad E.D.A. 10.0.6		Id: 1/3

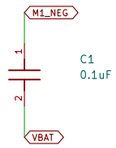
ESP-FLY | Bốn mạch điều khiển động cơ chổi than

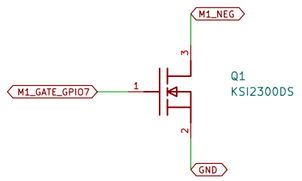
Tái dựng từ Gerber Max Imagination, bản 331d8f50 | 4 lớp | dày 1,2 mm

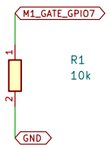
M1 / GPIO7



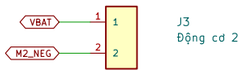


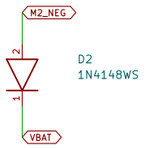


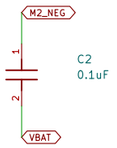


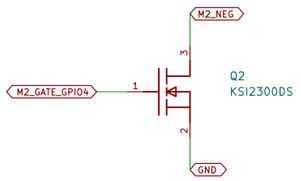


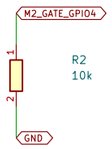
M2 / GPIO4



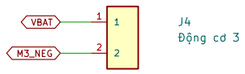


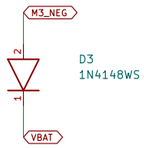


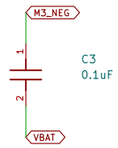


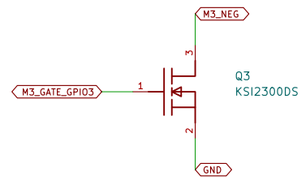


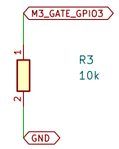
M3 / GPIO3



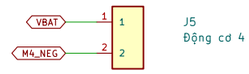


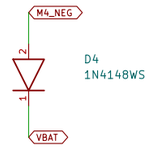


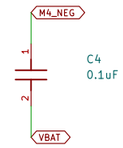


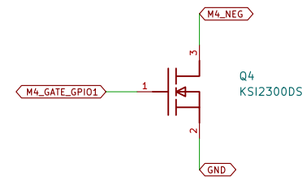


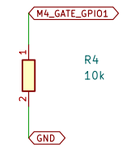
M4 / GPIO1











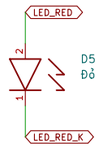
J2–J5 tương ứng các pad hàn P1–P8 trong thiết kế gốc, chia thành bốn đầu nối hai chân.
Chân 1: VBAT (+). Chân 2: cực âm động cơ qua MOSFET.
Thiết kế gốc không có điện trở nối tiếp ở cổng MOSFET.
Xem sơ đồ đấu dây bản chổi than và cấu hình động cơ trong firmware để xác định vị trí, chiều quay.

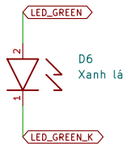
Gerber nguồn 331d8f50; rà soát DRC trước khi chế tạo
Thiết kế gốc: Max Imagination

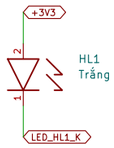
Sheet: /Động cơ/		
File: Động-cơ.kicad_sch		
Title: ESP-FLY | Bốn mạch điều khiển động cơ chổi than		
Size: A3	Date: 2026-09-25	**Rev: Tái dựng 1**
KiCad E.D.A. 10.0.6		Id: 2/3

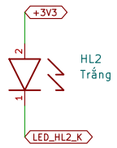
ESP–FLY | Đèn trạng thái và đèn định hướng

Tái dựng từ Gerber Max Imagination, bản 331d8f50 | 4 lớp | dày 1,2 mm

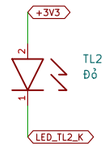


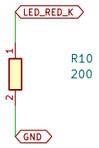


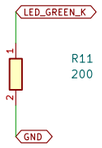


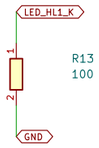


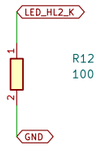


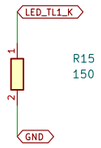


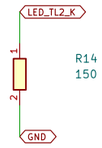












Đèn báo đỏ / xanh: GPIO43 và GPIO9.
HL1/HL2: đèn trước màu trắng, 100 Ω. TL1/TL2: đèn sau màu đỏ, 150 Ω.
Cả bốn đèn định hướng được cấp nguồn liên tục từ +3V3.

Gerber nguồn 331d8f50; rà soát DRC trước khi chế tạo
Thiết kế gốc: Max Imagination

Sheet: /Đèn LED/
File: Đèn–LED.kicad_sch

Title: ESP–FLY | Đèn trạng thái và đèn định hướng

Size: A3	Date: 2026–09–25	**Rev: Tái dựng 1**
KiCad E.D.A. 10.0.6		Id: 3/3